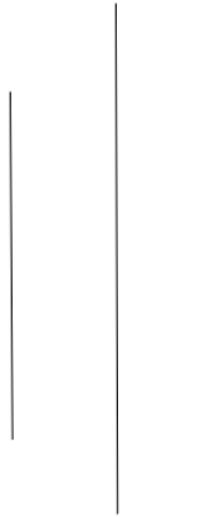


Levelling & Traverse surveying

Combine Report



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levelling

Levelling is a crucial surveying technique employed to assess height variations between different locations, thereby ensuring precision in engineering and construction endeavors.

This method is essential for generating accurate elevation data necessary for the construction of foundations, drainage systems, and road inclines. Additionally, it aids in identifying ground shifts and structural changes, which are vital for maintaining long-term safety and operational integrity. Commonly utilized techniques include the rise and fall method and the height of collimation method. Although levelling necessitates the use of specialized instruments and skilled professionals, its advantages significantly surpass the associated challenges, as it is instrumental in averting design flaws and facilitating effective infrastructure development. ((Stiros, 2012)

Purpose of Levelling

- ❖ Designing Level or sloped surface such as railways, roads, and drainage system. (Ghilani & Wolf, 2012)
- ❖ Ensuring Building Foundation are at correct height.(Uren & Price, 2018)
- ❖ Setting out construction works accurately. (Schofield & Breach, 2007)
- ❖ Monitoring Ground Movement or subsidence. (Stiros, 2012)
- ❖ Creating Contours maps and topographic profile. (Ghilani & Wolf, 2012)
- ❖ Planning flood defense and water flow in infrastructure. (Reynolds, 2011)

Explanation:

1. Designing Level or sloped surface.

Levelling is a critical process in the design of both flat and inclined surfaces, including railways, roads, and drainage systems.

It is vital for achieving correct alignment, maintaining smooth gradients, and facilitating effective water drainage. Precise elevation data is indispensable

for engineers as it enables the planning of safe and functional infrastructure, thereby mitigating potential problems such as flooding or irregular surfaces.

2. Ensuring Building Foundation is at correct height

Achieving proper levelling is essential for establishing building foundations at the appropriate height, thereby creating a stable and uniform base for construction. Precise levelling helps avert structural complications, water drainage problems, and uneven settling, all of which can jeopardize the structural integrity and longevity of a building.

3. Setting out construction works accurately.

Leveling is crucial for the precise execution of construction projects, as it facilitates the transfer of design elevations from blueprints to the actual site. This process guarantees that structures are constructed at the appropriate heights, in accordance with engineering requirements.

By ensuring accurate levelling, the likelihood of construction errors is reduced, thereby enhancing quality control and enabling effective coordination throughout different phases of the building process, from excavation to the final touches. Such precision is vital for ensuring safety, stability, and adherence to design standards.

4. Monitoring Ground Movement

Levelling serves as an essential method for tracking ground movement or subsidence, enabling surveyors to observe even slight variations in elevation over time. Through the consistent measurement of reference points, it aids in recognizing structural alterations or land settlement that could signify geological instability or effects related to construction activities.

This data is crucial for evaluating risks, planning necessary interventions, and maintaining the long-term safety of infrastructure and adjacent regions. Precise levelling facilitates the prompt identification of potential hazards.

5. Creating Contours maps

Levelling plays a crucial role in the development of contour maps and topographic profiles, which illustrate the elevation and configuration of the land surface. By gathering precise height measurements at multiple locations, surveyors can create contour lines that link regions of identical elevation.

These maps are indispensable for the planning of construction initiatives, land use management, and the comprehension of terrain characteristics. Additionally, topographic profiles facilitate the visualization of elevation variations along a designated path, supporting infrastructure design and environmental assessments.

Advantages of Levelling (Kavanagh & Bird, 2000)

- Accurate elevation data
- Ensure structural safety
- Supports drainage and water flow design
- Monitors Land movement
- Improve project planning

Disadvantages of Levelling (Anderson & Mikhail, 1998)

- Time consuming process
- Accuracy depends on equipment and conditions
- Labor intensive
- Limited coverage area
- Susceptible to external factors

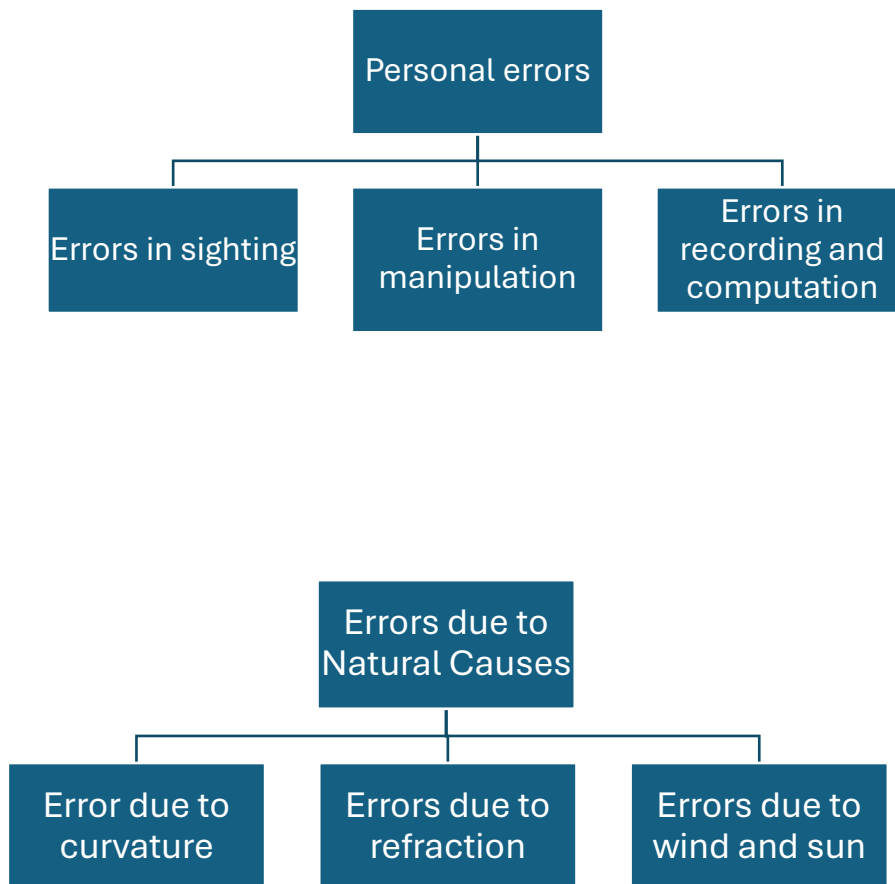
Uses of Levelling

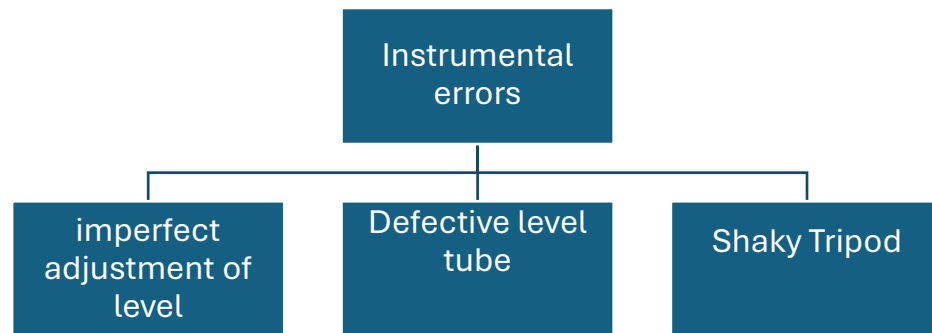
- I. Longitudinal Sections: It is used for preparing longitudinal section along the center line of proposed alignment of highway, canal, pipelines etc.
- II. Cross Sections: It is used for preparing the cross – section on either side of center line for the purpose of determining the lateral layout of the ground.

- III. Contouring: By levelling operation, elevations of various points and important features can be determined on contouring.
- IV. Setting out levels: Sometimes, a levelling survey in a small area for specific purposes requires R.L. To fix such elevation we can use levelling operation.
- V. Transferring R. L: Levelling is also used to transfer R. Ls of B.M to other points on ground

Errors in Levelling

The errors in levelling may be classified as:





Temporary Adjustment of Level

The adjustment which are made for every setting of a level, are called temporary adjustment of a level. These includes:

i. **Setting up the Level**

This operation includes fixing the instrument on the tripod and also levelling the instrument approximately by leg adjustment. The tripod legs are so adjusted that the telescope is at a convenient height and the tribrach is approximately levelled.

ii. **Levelling up**

After setting up the level, accurate levelling is done with the help of foot screw and by using plate levels. The objective of levelling up the instrument is to make its vertical axis is truly vertical.

iii. **Elimination of parallax**

If the image formed by the objective does not lie in the plane of the crosshair, there will be a shift of the eye. Such displacement of image is termed as parallax. For accurate sighting, the parallax is eliminated in two steps:

a. **Focusing the eye piece for distinction vision of cross hair:**

Following steps are involved:

- Direct the telescope either towards the sky or hold a sheet of a white paper in front of the objective.
 - Move the eye piece in or out till the cross hair appear distinct
- b. Focusing the objective so that the image is formed in the plane of cross hair
- Direct the telescope towards the levelling staff
 - Turn the focusing screw till image appear clean and sharp
 - The image so formed must be in the plane of cross hair.

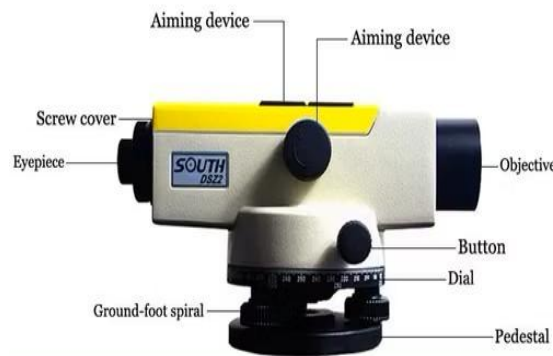


Fig: parts of auto level

Method of Levelling

1. Rise and Fall method
2. Height of collimation method

Rise and Fall method

The Rise and Fall Method is a conventional approach employed in differential leveling to ascertain the elevation difference between various points. This technique is frequently utilized in surveying and construction to evaluate the vertical distance, or height variation, between distinct locations on the Earth's surface.

How does this method work?

In this method, the levelling process involves recording the backsight (B.S) and foresight (F.S) reading to establish height differences between the consecutive points. The total height difference is calculated using a series of adjustments, typically involving multiple set up of the instrument.

Advantages of Rise and Fall Method

- I. Simple to perform
- II. Reliability
- III. Cost-Effective

Disadvantages of Rise and Fall method

- I. Cumulative Errors
- II. Labor intensive
- III. Requires good conditions

Height of collimation method

It refers to the elevation of the line of sight of the levelling instrument above a known reference point, usually called as benchmark or any point with a known reduced level (R.L)

How does this method work?

1. Setting up the levelling instrument at desired location
2. Take a backsight reading on a point with a known elevation or a benchmark.
3. And, then take an intermediate reading and take foresight before changing the station.

4. Calculate the height of the collimation using:

$$\text{HOC} = \text{RL of the point} + \text{B.S}$$

5. Determine the RL of other points.

$$\text{RL} = \text{HOC} - \text{F. S or I.S}$$

Advantage of Height of Collimation method

- I. Simple and Quick calculations
- II. Ideal for Long Runs
- III. Good for profile Levelling
- IV. Less prone to small reading errors

Disadvantages of Height of Collimation Method

- I. Difficult to detect errors
- II. No check on arithmetic accuracy
- III. Not ideal for detailed survey
- IV. Relies on consistent instrument setup

Comparison: Rise and Fall Method Vs Height of Collimation Method

Aspect	Rise and Fall Method	Height of Collimation Method
Principle	Calculates difference in height between points	Determines height of instrument line (collimation)
Calculations	Based on rises and falls between consecutive points	Based on height of collimation and foresight readings
Error Checking	Easy to detect and correct errors	Difficult to detect computational mistakes
Accuracy	More accurate for detailed surveys	Slightly less accurate over long distances
Suitability	Best for precision work	Best for quick RL (Reduced Level) calculations
Field Book Format	Requires BS, IS, FS, Rise, Fall, RL columns	Requires BS, IS, FS, HOC, RL columns
Ease of Use	More complex and time-consuming	Simpler and faster
Use Cases	Engineering surveys, construction projects	General levelling, quick height assessments

Difference Between Rise and fall and height of collimation

Height of collimation Method	Rise and Fall Method
It is more rapid and saves considerable time and labor	It is laborious as the staff reading of each station is compared to getting a rise or fall.
It is well adopted for reduction of levels for construction work such as longitudinal or cross-section levelling operation	It is well adapted for determining the difference in levels of two points where precision is required
There is no check on reduction of R.L if intermediate stations.	There is a complete check on the reduction of R.Ls. of intermediate stations
There are only two arithmetical checks, i.e. the difference between the sum of the back sights and sum of fore sights must be equal to the difference in R.L. of the last station and first station	There are three arithmetical checks, i.e. the difference between the sum of back sights and the sum of the fore sights must be equal to the difference between the sum of rises and sum of falls as well as it must also be equal to the difference in R.Ls. of the last station and first station
Errors, if any in intermediate sights are not detected.	Errors in intermediate sights are noticed as these are used for finding out rises and falls



Fig: Auto Level

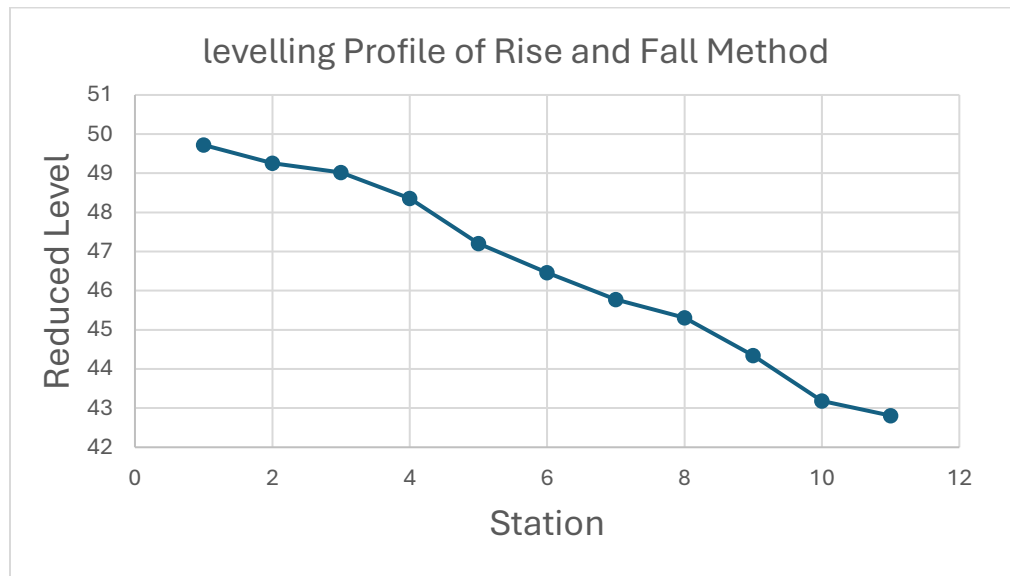


Fig: Level Staff



Fig: Tripod Stand

Leveling Profile for Rise and Fall Method



What does the graph show?

- The line slopes downward, meaning the ground surface is gradually decreasing on elevation.
- RL starts at 49.722m and decreases to about 42.809m, showing a total fall of 6.913m.
- The curve is mostly smooth, showing a consistent gradient or slope

Calculation for Rise and Fall Method

SN	BS	IS	FS	Fall	RISE	RL
1	1.172					49.722
2	1.072		1.635	0.463		49.259
3		1.315		0.243		49.016
4	1.16		1.973	0.658		48.358
5		2.31		1.15		47.208
6	1.209		3.06	0.75		46.458
7		1.89		0.681		45.777
8		2.36		0.47		45.307
9	0.47		3.32	0.96		44.347
10	1.04		2.635	2.165		42.182
11			1.413	0.373		41.809
SUM	6.123		14.036	7.913		

Arithmetic check for the given calculation:

$$\sum Bs - \sum FS = \sum Rise - \sum Fall = \text{Last RL} - \text{First RL}$$

$$7.123 - 14.036 = 0 - 6.913 = 41.809 - 49.722$$

$$-7.913 = -7.913 = -7.913$$

Hence, the arithmetic calculation shows everything is correct

Summary for the Data taken and calculation

To collect this data first we took a backsight of a known reduced level and then we changed station, before changing the station we have taken a Foresight. After that we took Intermediate sight. In the same way we have taken the data and changed the station as required. We changed the station 6 times during the levelling of our site.

The starting point of our levelling was the hive café with a reduced level of 49.722m and the last station of our levelling was the closest point of the traverse which is near the entrance of University of Roehampton Library. And the Reduced level we have calculated is 42.089m.

This shows that the area we perform a levelling is quite sloppy going downward from our first station. And we can observe that there is no rise while we calculate the data taken from the site. We had only observed the fall in the calculation table above.

Rise Fall and RL Calculation

This method compares the BS, IS, and FS reading between the stations:

For Each row:

Rise/ Fall = Previous reading – Next Reading (If positive = rise and if Negative = Fall)

Let's break it down to understand the calculation:

For example,

Row 1

BS = 1.172

Next FS = 1.635

Difference = $1.172 - 1.635 = -0.463\text{m}$ (Fall)

In the same way all the others fall are calculated.

For the Reduced Level,

We calculated RLs based on the rise/ fall from starting RL (Which is given: 49.722m at station 1)

For example,

Station 1 (starting point)

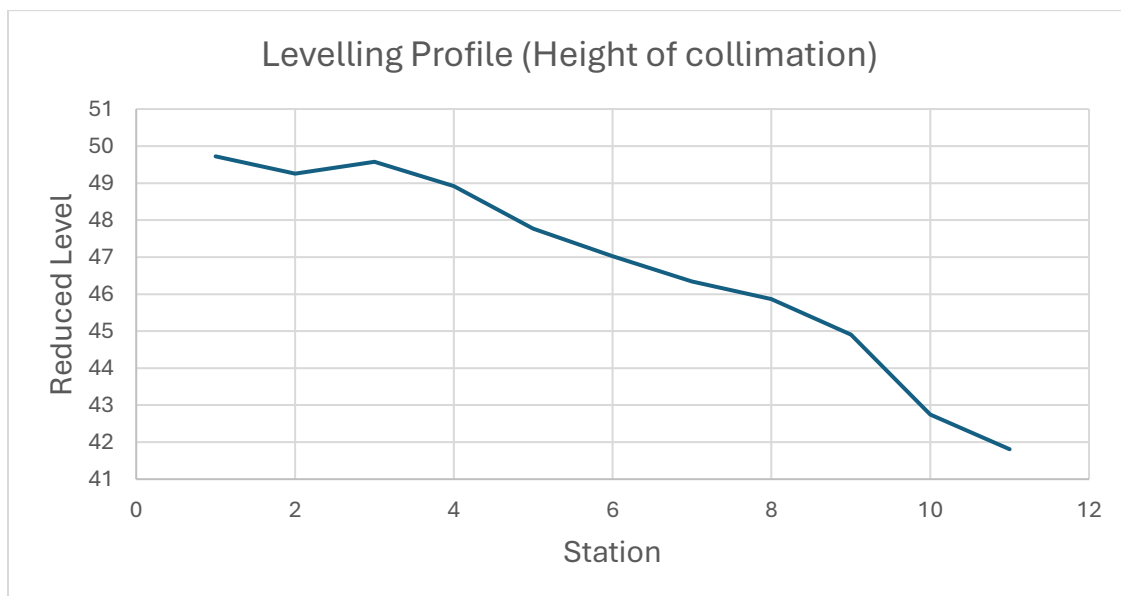
RL= 49.722m

Station 2:

Fall = 0.436m RL = $49.722 - 0.463 = 49.259\text{m}$

In the same way all the RL in the above table is calculated and the RL of last station is calculated 42.089m

Levelling Profile for height of Collimation Method



Calculation for Height of Collimation Method

SN	BS	IS	FS	HI	RL
1	1.172			50.894	49.722
2	1.072		1.635	50.331	49.259
3		1.315			49.579
4	1.16		1.973	50.081	48.921
5		2.31			47.771
6	1.209		3.06	48.23	47.021
7		1.89			46.34
8		2.36			45.87
9	0.47		3.32	45.38	44.91
10	1.04		2.635	43.785	42.745
11			1.413		41.809
SUM	6.123		14.036		

Arithmetic check for the given calculation:

$$\sum BS - \sum FS = \sum Rise - \sum Fall = \text{Last RL} - \text{First RL}$$

$$7.123 - 14.036 = 0 - 6.913 = 41.809 - 49.722$$

$$-7.913 = -7.913 = -7.913$$

Height of collimation and RL Calculation

1. Calculate Height of Instrument

$$H.I = R.L + B.S$$

At each new instrument setup, HI is calculated using the RL of the point where Back Sight is taken:

For example,

- RL = 49.722
- BS = 1.172

$$\text{Height of Collimation} = \text{RL} + \text{BS}$$

$$= 49.722 + 1.172 = 50.894\text{m}$$

Calculate the RL for the subsequent Points

$$\text{RL} = \text{HI} - \text{IS or FS}$$

Depending on whether it's an intermediate or foresight reading.

For example,

- HI = 50.894m
 - FS = 1.635m
- $$\begin{aligned}\text{RL of a point} &= 50.894 - 1.635 \\ &= 49.529\text{m}\end{aligned}$$

This continues for each point. Every time there's a BS; a new HI is computed using the RL at that point.

Personal portfolio of each member

1. Hanan Hanid

I play a crucial role in the levelling survey project as the team's data analyst responsible for processing and interpreting the raw field data. While I could not attend the onsite measurement due to the fever. My role was essential in ensuring the accuracy and integrity of the results. After I received the field data and staff reading from my group, I carefully applied the Rise and Fall method to calculate the Reduced level at each station.

I was responsible for identifying and correcting inconsistencies, ensuring all the arithmetic checks were satisfied, and organizing the results into clear tables and profiles. I also contributed by plotting the levelling profile graph to visualize the terrain, allowing the team to understand elevation changes along the survey route. My attention to detail and analytical skills supported the team's final report and contributed to the overall success of the project.



2. Chandan Mahara

I joined the levelling survey at its final phase, stepping in when only the last two stations remain to be completed. Despite entering at a later stage, I played an important role in ensuring the survey accuracy and completion. Upon arrival, I assisted in recording the final field reading and took the responsibilities for computing the reduced level using the height of collimation method.



Beyond the calculations, I also prepare a level profiling, visualizing the terrain's elevation across the surveyed line. My contributions helped finalize the levelling table, confirm the consistency of the instrument's line of sight, ensuring the closing RL match expectations. The visual profile helped the team to confirm that the terrain data was accurate. My ability to join late yet meaningfully contributes to both the technical calculations and results of data reflects his focus and understanding of levelling principles.

3. Bibek Jabegu

I played a crucial role in the leveling survey as a key team member from the project's inception. My main responsibility involved operating the auto level instrument, where I made significant contributions to the survey's accuracy through effective management of equipment setup and on-site coordination. I oversaw the instrument setup at four distinct stations, including Station 6, ensuring that the auto level was properly leveled, focused, and aligned for accurate readings. My knowledge of the instrument and techniques to reduce observational errors were instrumental in maintaining consistency across all stations. Additionally, I provided support as a staff holder while utilizing the auto level. My role as both an instrument



operator and a data recorder show my adaptability and dependability on site. I made sure all the field data was accurately recorded, setting the stage for precise calculations down the line. My early participation and consistent input played a key part in creating a seamless workflow for the whole survey team.

4. Rochak Lamsal (Rochu)

I spearheaded the initial stage of the levelling survey, taking charge of the operation of the auto level instrument from the starting point to Station 6. As the first individual to set up the equipment and commence measurements, I played a pivotal role in establishing the foundational reference and ensuring that the readings were both accurate and aligned with the project's levelling criteria.

Utilizing the auto level, I meticulously recorded Back Sight (BS), Intermediate Sight (IS), and Fore Sight (FS) readings at various stations, effectively guiding the team through the initial segment of the survey. My proficiency with the equipment and surveying methodologies facilitated consistent and precise data collection, which became the basis for all subsequent measurements. Following Station 6, I contributed to the field by holding the levelling staff and continued to assist with reading and documenting data. My involvement was essential in preserving the accuracy and continuity of the survey throughout its various phases.



Conclusion

A levelling survey carried out at the University of Roehampton campus showed a steady decline from the Hive Café to the Library entrance, confirming a total elevation drop of about 6.913 meters. The team utilized both the Rise and Fall method and the Height of Collimation method to accurately determine the

Reduced Levels (RLs) at each point, with thorough arithmetic checks ensuring the reliability of the data. These levelling methods were crucial for mapping changes in the terrain and will be vital for future projects involving drainage, structural design, or landscaping enhancements in the area.

The success of the survey was due to the teamwork of all members, who participated in data gathering, equipment management, calculations, and profiling, ensuring a comprehensive and accurate survey. Even with challenges like illness or tardiness, each member played a significant role, demonstrating excellent collaboration and technical skills. This project not only showcased the practical uses of levelling but also emphasized its vital importance in achieving safe and precise engineering results.

Additionally, the experience deepened the team's knowledge of levelling theory and hands-on fieldwork. Working with real surveying tools such as the auto level, tripod, and staff, along with calculating RLs using various methods, helped team members understand the intricacies and accuracy needed in construction surveying. It also highlighted the necessity of being detail-oriented, as minor observational mistakes could greatly impact the final outcomes.

In summary, this project offered valuable practical experience in levelling, reinforcing key civil engineering skills like precision, collaboration, and data analysis. The results not only met expectations but also illustrated the significance of levelling in ensuring safe and well-organized construction. This foundational experience lays the groundwork for future endeavors.

Note: Some of the things are taken from my diploma book which I didn't find on the internet so I could not provide a references.

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